

FD-LSTM: A Fuzzy LSTM Model for Chaotic Time-Series Prediction

¹Milad Keshtkar Langeroudi, ²Mohammad Reza Yamaghani, ³Siavash Khodaparast
^{1,2} Department of Computer Engineering, Lahijan Branch, Islamic Azad University, Lahijan, Iran
³ Department of Physical Education, Lahijan Branch, Islamic Azad University, Lahijan, Iran

Abstract

The main issue of time-series prediction is to determine the grade of uncertainty in knowledge, with its essential vagueness and haziness in complex problems. In this study, a Deep Fuzzy LSTM architecture has been proposed to handle the high-order uncertainty associated with time-series applications. The LSTM and type-2 fuzzy logic combination aim to make a more transparent, interpretable, and accurate predictive system. The experiments of this study with real data contain global standard and real-valued benchmarks, including Mackey-Glass, Sunspot, and EPL Premier League seasonal datasets. The obtained performance shows the superiority of the proposed Fuzzy-Deep model in predicting time-series with an average AUC=0.96 in Sunspot, 0.93 for Football Match time-series, and 0.95 on a chaotic equation of Mackey-Glass.

Index Terms- Time-Series Prediction, Deep Learning, LSTM, Type-2 fuzzy system

Introduction

Prediction of a times series has a significant act in numerous applied discipline areas in real-world applications. The scientific and engineering communities have noticed the importance of time-series dynamical characteristics, the study, and the forecast of disordered and chaotical time series. Real-life glitches and problems necessitate misuse of architecture that permits coping with different levels and types of uncertainty. Also, a time-series are omnipresent, seeming in nearly every study and research where data and information are studied and analyzed. Though their routine analysis entails superior statistical techniques and concepts, so without which flawed interpretations, inferences, and decisions may all too willingly be drawn. Once a time series is inspected and observed at quarterly or yearly intervals, its annual seasonal order (pattern) is a vital feature [2]. Another important characteristic of time series is the non-appearance of any continued increase or failure in the level of each series over the observation duration. This feature enables time series to predict from the physical nature of every series, which is called Stationary time series [3].

The nature of time-series and its chaotic form comprises big data size, high dimensionality, high order uncertainty, and is required to bring up-to-date. Additionally, the pattern and data of time-series, considered by its continuous and numerical structure, are continually measured as a whole instead of the discrete numerical field [4]. The growing use of time series has started an inordinate pact of study and development efforts in pattern recognition and data mining.

Research Motivation

The main objective of this study is to present a novel fuzzy deep learning model for chaotic time series prediction. In this work, the proposed model aimed to overcome the drawbacks of the existing methods and present a robust, reliable, and accurate model for predicting chaotic time-series data through type-2 fuzzy logic and LSTM models. On the other hand, precise and reliable prediction can help reasonable strategies and assist specialists in planning the best policies to model an event in uncertain circumstances.

Computational intelligence methods, such as fuzzy logic and deep neural networks, are robust models to solve real-world problems. In many dynamic and complex problems, statistical attributes frequently change over time. Recurrent neural networks (RNN) are suitable to model dynamic high-dimensional and nonlinear state-space systems. Nevertheless, the RNN cannot model long-term dependencies in temporal data. Its learning using gradient descent is a complex and challenging task. Long Short-Term Memory (LSTM) networks were introduced to overcome the RNN issues, but coping with uncertainty is still a significant challenge for the LSTM models. Fuzzy logic, especially Type-2 Fuzzy logic, is a robust and capable model to cope with high-order uncertainties associated with non-stationary time-dependent features.

On the other hand, the fuzzy inference rules make feature extraction more human interpretable under uncertainty. Utilizing the hybrid intelligent models brings the advantages of fuzzy logic and deep learning approaches in modeling the uncertainty and extracting useful high-level features from ambiguous data.

Research Background and Literature Review

Fuzzy time-series prediction has become a significant area that advances too much attention from researchers. Data mining and pattern recognition researchers have studied those areas in ample detail for different domains of uncertain time series. In the background of time-series prediction, the fundamental problem is how an automatic system can represent the time series and handle the associated uncertainty. Prediction is an effective way to reveal and visualize the structure of uncertain data [5]. So, choosing an appropriate measure to assess the similarities/dissimilarities inside the data becomes required. It necessarily displays an essential impact on the prediction results. This selection would be based upon the natural characteristic of time series and the application itself [6]. In this part of this study, various approaches have been reviewed. As soon as historical data are etymological values, the conventional neural time series cannot cope with the prediction problems [7]. Another imperative interrogation is around the variables applied in a fuzzy time-series prediction. Several classic fuzzy time-series models first apply one variable in the process of displaying, specifically a Type-1 fuzzy time-series prediction system. The variable applied in the Type-1 fuzzy system is also known as Type-1 observation. Though, for a few high-order and chaotic models, the modification of the value of a variable is not one relating to its own rules but also pretentious through other aspects and factors. A Type-2 fuzzy time-series prediction system is presented in [8] to benefit the relevant factors. Later, some other research has led to the development Type-2 fuzzy time-series prediction system. In [9], a Type-2 predictive classifier is described as the optimal prediction truth through predicting the optimal membership functions. Later, this method is protracted to the fourth-order model.

Type-2 Fuzzy Logic and Time-Series

Lotfi Zadeh presented the first idea of a type-2 fuzzy set in 1975 as an extension of a standard fuzzy set known as a type-1 fuzzy set. On the other hand, a type-2 fuzzy set is considered via a fuzzy membership function; for example, the membership grade intended for respectively composed of a set is a fuzzy set such as in $[0, 1]$, dissimilar a type-1 set which the membership grade is a crisp value in $[0, 1]$. Type-2 fuzzy sets take a broad view form of type-1 fuzzy sets. They can cope with uncertainty in complex environments. The membership function (MF) of type-2 fuzzy logic is called the primary MF or Footprint of Uncertainty (FOU). The FOU is the two-dimensional support of $\tilde{A}$, follow as:

$$FOU(\tilde{A}) = \{(x, u) \in X \times U \mid \tilde{\mu}_A(x, u) > 0\} \quad (1)$$

$$\mu_{A(x)} = N(x, m, \sigma_2) \text{ for UMF} \quad (2)$$

$$\mu_{A(x)} = N(x, m, \sigma_1) \text{ for LMF} \quad (3)$$

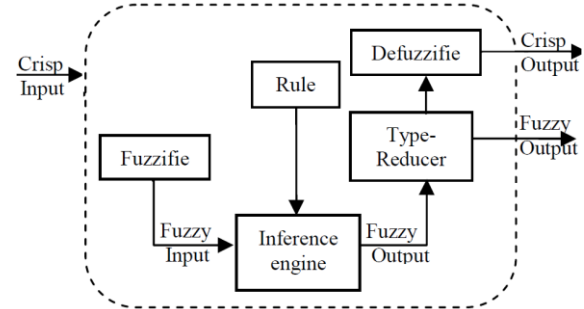


Fig.1 Scheme of a standard Interval Type-2 Fuzzy System [3]

RNN and LSTM

Recurrent Neural Networks, known as RNNs are artificial neural network models that are compatible with pattern mining and classification problems through inputs and outputs are the order of the sequences. The role of emerging approaches for mapping the sequences to sequences is demonstrated thru errands such as pattern recognition and time-series prediction. An RNN signifies a sequence with a high-dimensional vector, which we call the hidden state, of a fixed dimensionality that includes innovative explanations using a complicated nonlinear function. They are more susceptible than would implement subjective memory-bounded calculation. Consequently, it would likely be constructed to attain a non-trivial recital on demanding and complex sequence procedures. Since recurrent nets budding has not been realized, approaches that handle the trouble of train procedure in RNNs and their excessive status. RNNs are inordinate for processing sequences of data, for example, a dataset of pollution particles or temperature over time

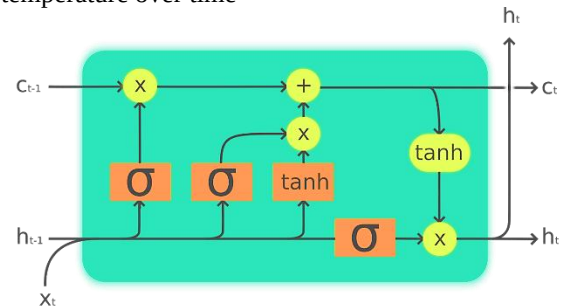


Fig.2: Structure of a recurrent LSTM network [4]

The figure shows a single neural network node (A) with an input (X) and output (h). However, there is a twist. Node A is connected to itself with a looping connection. It can picture the loop by unrolling it. The

right cross of the figure demonstrations the unrolled network. Each input is now at a precise point in time. It starts by feeding X_0 into the network, then X_1 , then X_2 , and so on. The flat connectors provide short-term memory by feeding the network state into the next iteration. On the other hand, A LSTM network is an explicit type of RNN construction intended to model temporal sequences and the long-time-series and ranges dependencies more accurately than conventional recurrent networks. In LSTM structure, every memory cell's inner construction assurance relentless error within its Constant Error Carrousel or CEC. It characterizes the origin designed for linking very long-time lags. The two gates (units) acquire to close and expose admission to a fault within each memory cell of CEC.

Proposed Model: Fuzzy-Deep LSTM (FD-LSTM)

In figure 3, the complete structure of the proposed architecture has been revealed.

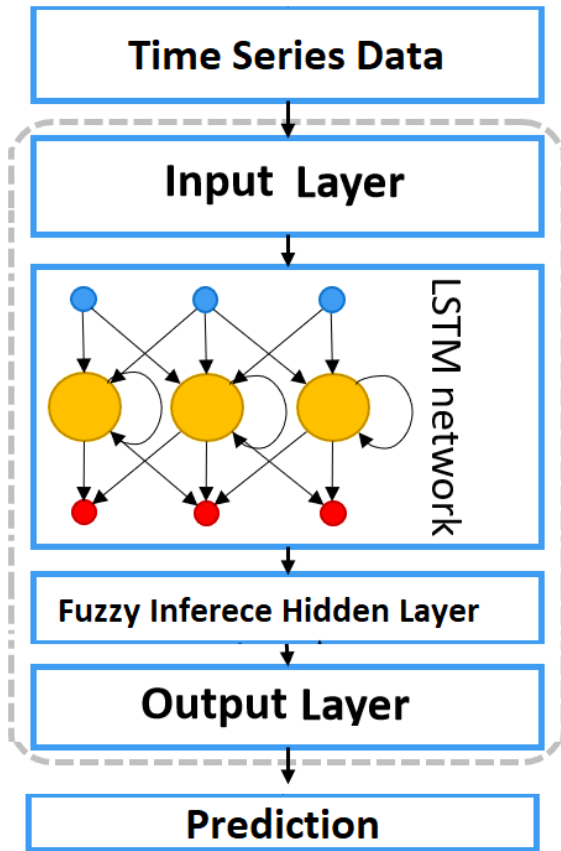


Fig3. the block diagram of the proposed architecture

The proposed Fuzzy-Deep LSTM (FD-LSTM) model stylishly handles the vanishing gradients delinquent based on memory units. These linear units have a self-construction of strength and a couple of auxiliaries

gating units that handle the current information from the unit.

First Layer: Input Layer

The first layer is the input layer, which directly controls the primary data. The output of this input layer feeds the following hidden or deep layers. The input variables of the proposed model can be defined as the inputs of the proposed model:

$$p \begin{cases} R_p^{j_1, j_2}: \text{if } p_t = \tilde{F}_p^{j_1} \text{ and } p_{t+1} = \tilde{F}_p^{j_2} \\ \text{then } y_p = [\tilde{c}_p^{j_1, j_2}, \tilde{c}_p^{j_1, j_2}] \end{cases} \quad (4)$$

Where $\tilde{F}_p^{j_1}$ and $\tilde{F}_p^{j_2}$ are the inputs of interval type-2 fuzzy sets.

This layer fuzzifies the which map a nonlinear function in discrete time which defined as follows:

$$y_{(k)} = \varphi[U_r(k)] \quad (5)$$

Where φ is a nonlinear relation for calculating uncertainty and $U_r(k)$ is also a state vector defined as:

$$U_r(k) = y_{(k-1)}, \dots, y_{(k-n_y)}, U_{(k)}, \dots, u_{(k-n_u)}]^T = [u_{r_1} \dots u_{r_m}]^T \quad (6)$$

Where $u_{(k)}$ and $y_{(k)}$ are the amounts of input and output signal of the system and n_y indicates the delays in the input of the FD-LSTM system. Also, m represents the number of elements in u_r within $u_{(k)}$.

Second Layer: Cell States, Inference, and the Hidden Layers

This layer's main function is to plot the entire input sequence into a time vector. This layer consists of K -nodes, representing the current rule's strength based on type 2 fuzzy inference engine. To define the FD-LSTM part, let N be the number of memory units of the model. In every time-step t , which is the current time, the network keeps a set of vectors, whose growth is governed by the resulting equations:

$$i_t = \sigma(W_{ix}x_t + W_{im}m_{t-1} + W_{ic}c_{t-1} + b_i) \quad (7)$$

$$f_t = \sigma(W_{fx}x_t + W_{fm}m_{t-1} + W_{fc}c_{t-1} + b_f) \quad (8)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot g(W_{cx}x_t + W_{cm}m_{t-1} + W_{cc}c_{t-1} + c) \quad (9)$$

$$o_t = \sigma(W_{ox}x_t + W_{om}m_{t-1} + W_{oc}c_t + b_o) \quad (10)$$

$$m_t = o_t \odot h(c_t) \quad (11)$$

$$y_t = \varphi(W_{ym}m_t + b_y) \quad (12)$$

where σ is the logistic sigmoid function, the W mean weight matrices. e.g., W_{ix} is a matrix of fuzzy weights

from the input cell to the output gate, the b term signifies the bias vectors and i, f, o and c are individually the input gate, forget gate, output gate and cell activation vectors, all of them are the same size as the cell output activation vector m , $\odot$ is the element-wise product. The proposed structure can be modified, retrained, and retrained according to the dimensions of the time series data set. The midpoint of the memory cell is a humble linear component with a self-recurrent association that weighs a fuzzy value between 0 and 1. In the absence of other inputs, this joining upholds the current state of the cell. The cells also obtain input from the input unit and the other gates. So, let us to describe the fuzzy part and the squashing function used on all gates in the fuzzy range [0,1]:

$$f(x) = \frac{1}{1+e^{-x}} \quad (13)$$

The input to the cell (net_c) uses the following equation:

$$net_c^y(t) = \sum_m W_{\phi jm} y^m(t-1) \quad (14)$$

The input to the cell passes a nonlinear function $g(x)$, a logistic function with a scale fuzzy value from -3 to 3:

$$g(x) = \frac{4}{1+e^{-x}} - 2 \quad (15)$$

Moreover, for defining the Antecedent MFs, the chief MFs for every antecedent are described through Gaussian distribution with uncertain income, which is shown in the following equation:

$$\mu_k^i(X_k) = \exp \left[-\frac{1}{2} \left[\frac{X_k - m_k^i}{\sigma_k^i} \right]^2 \right] \quad (16)$$

Where $m_k^i \in [m_{k1}^i, m_{k2}^i]$ is the ambiguous and uncertain mean, with $k=1,2$ which are the number of antecedents and $i = (1,2,3,4, \dots, n)$ which are the number of fuzzy rules, and σ_k^i is the standard deviation.

Third Layer: Output Layer

The fuzzified information needs to be reduced in type-1 fuzzy and converted to a crisp value. This process receives the inputs from the hidden layers of the last part of the architecture, which are interpreted and determined by the fuzzy inference system. Then, all the values are non-fuzzy and converted into a two-dimensional array instead of modeling. Finally, the obtained results are used as the final modeling of time patterns to make decisions and make predictions based on the characteristics received from the previously hidden layers. In order to apply a type-reduction process, a Karnik-Mendel algorithm has been used as follows:

$$y(k) = \frac{\sum_{i=1}^k \bar{y}^i \underline{\mu}_{\tilde{G}_{out}}^i + \sum_{i=k+1}^K \bar{y}^i \bar{\mu}_{\tilde{G}_{out}}^i}{\sum_{i=1}^k \underline{\mu}_{\tilde{G}_{out}}^i + \sum_{i=k+1}^K \bar{\mu}_{\tilde{G}_{out}}^i} \quad (17)$$

where k is an integer in the range $[1, K-1]$ and K represents the number of discrete points.

Empirical Analysis

Deep long short-term memory architectures known as Deep-LSTMs are built by assembling numerous LSTM layers. Also, we need to clarify that LSTMs are even now a deep system in the intellect. They can be measured as a feed-forward neural net unrolled in time series where every layer bonds the same system parameter. Also, the model inputs go through numerous nonlinear layers, as in Deep Neural Nets (DNN).

However, the structures from a given time prompt are only processed by a single nonlinear layer before causative the output for that time instant. Consequently, the depth in the deep LSTM model has a supplementary sense. The input to the network at a given time step goes concluded multiple LSTM layers in accumulation to propagation determined time and LSTM network layers. Another benefit of Deep LSTM offer over standard LSTM is that it can make improved applying of parameters by allocating them over the space concluded numerous layers.

For instance, rather than growing the memory size of a standard model by a factor of two, one can have four layers with approximately the same number of parameters, known as LSTMP networks, within a projection part associated with its architecture which additional tuned speed and performance of LSTM by adding a projection layer. On the other hand, like deep LSTM, a deep LSTMP has multiple layers. Each with a separate recurrent projection layer is fixed. LSTMP permits the model's memory to be enlarged autonomously from the output layer and recurrent connections. Still, growing the memory size makes the model more disposed to overfitting by memorizing the input sequence data.

This type of neural net generalizes better to unseen examples with increasing depth. The depth makes the architecture harder to over-fit to the training data. Meanwhile, the inputs to the network need to go over many nonlinear functions. The idea of fuzzy time series addressed only one feature first-order model.

Performance Evaluation

In this section, three different scenarios have been applied to guarantee the fairness of the proposed model, which are:

- 1- Mackey Glass Chaotic Time-series (Global Benchmark)
- 2- Sunspot Time-series (Standard Dataset)
- 3- EPL Football Match Time Series (Real-world Dataset)

For the prediction experiment in each scenario, we set the time window of the input sequence to prepare the datasets for the proposed FD-LSTM model. The main indexes are used to measure the error analysis of the proposed. We used the mean absolute error (MAE), the root mean square error (RMSE), and the Pearson correlation coefficient, which is denoted with p , where d_i is the error vector calculated, where N , y , and $\hat{y}$ denote the number of outed data, actual outcomes, and network output, respectively which are defined as follows:

$$RMSE = \sqrt{\frac{\sum_{i=1}^N d_i^2}{N}} \quad (18)$$

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_i - \hat{y}_i| \quad (19)$$

$$p = \frac{\sum_{i=1}^N (y_i - \bar{y})(\hat{y}_i - \bar{\hat{y}})}{\sqrt{\sum_{i=1}^N (y_i - \bar{y})^2} \sqrt{\sum_{i=1}^N (\hat{y}_i - \bar{\hat{y}})^2}} \quad (20)$$

On the other hand, reliable statistical evaluation of analytic performance in general and specifically ROC curve analysis was conducted to calculate the proposed model's performance. The confusion matrix was calculated to describe the performance of the proposed approaches. A ROC curve analysis was directed to evaluate the performance of the proposed model in different datasets. The data sets were separated into three parts; training, validation, and test. For this purpose, 10-Fold cross-validation was used to validate the results and make the classifier more reliable. The concept is to arbitrarily split the data into K (10) equal-sized parts. It leaves out part k , proper the approach to the other $K-1$ parts (joint), and then gets the prediction for the left-out K -th part. Intending to evaluate the precision of the passive samplers, we used real-time data and a validation set for 15% of 10-years samples taken at the reference sites all over the decade. Collected data were organized at all reference sites during the three periodic campaigns.

Specificity: The prospect of the test finding the correct class among all classes:

$$\frac{TN}{TN + FP} \quad (21)$$

Accuracy: The fraction of correct test results:

$$\frac{TP + TN}{TP + FN + TN + FP} \quad (22)$$

Precision: Precision or positive predictive value:

$$\frac{TP}{TP + FP} \quad (23)$$

Sensitivity (Recall): Hit rate

$$\frac{TP}{Total\ Positive} \quad (24)$$

Also, a left tailed t-test has been applied to denote the proficiency of the proposed model and its robustness. The null hypothesis is defined as $H_0 = \mu_i > \mu_j$ and $H_1: \mu_i < \mu_j$, where μ_i and μ_j are the means of FDLSTM and LSTM for 100 different runs with the 10-fold cross-validation technique as follow as:

$$\mu_i = \frac{1}{10} \sum_{k=1}^{10} AUC_j \quad (25)$$

Experimental Results

In this part, four mentioned time series were applied to evaluate the proposed model's capability to predict uncertain time series in both short and long-range time series. The first test is the Mackey-Glass chaotic time-series, a gold dataset for prediction problems, especially in high-order and chaotic situations. The second dataset is sunspot which is a standard dataset in time-series problems and their applications. The third dataset are real data which are collected from EPL premier league in British football matches. Also, the detailed of the experiment settings described as follows:

- CPU: Intel Core i7 10th gen 10750h
- GPU: RTX306
- Memory: 16 Gb DDR4
- OS: Windows 10 X64 Ultimate
- Matlab R2021b, PyTorch

Case Study 1: Mackey-Glass chaotic time-series

The proposed method was complemented by a Deep Type-2 Fuzzy LSTM (FD-LSTM) to obtain a newfangled predictor of the prediction's procedure. It permits us to calculate uncertainties of predictions for noisy Mackey-Glass (MG) chaotic time-series. In Figure 4, a demonstration of the chaotic attractor is aimed at this study. The test of the proposed model is a simulation of time-series data using the following form of the MG nonlinear delay differential equation:

$$\dot{x}(t) = \frac{0.2x(t-\tau)}{1+x^{10}(t-\tau)} - 0.1x(t) \quad (26)$$

Simulate the time series for 1200 samples using the following configuration:

- Sample time $t_s = 1$ sec
- Initial condition $x_0 = 1.2$
- $\tau = 20$
- $x(t - \tau) = 0$ for $t < \tau$.

Where λ_i is the infinite number of Lyapunov exponents.

Table-1: Applied proposed model on Mackey Glass

Evaluate Ratio	Noiseless	σN = 0.01	σN = 0.05	σN = 0.1
RMSE	0.012	0.011	0.078	0.109
rRMSE	1.091	1.167	3.890	9.267
MAE	1.834	2.050	2.331	2.869
p-Value	0.034	0.086	0.129	0.001

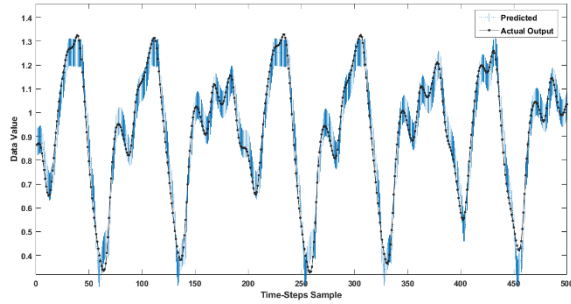


Fig 4: prediction in the MG chaotic time-series with a white noise.

Case Study 2: Sunspot Time-Series

Sunspots are temporary singularities on the Sun's photosphere that seem darker than the surrounding areas. They are areas of abridged surface temperature caused by the attention of magnetic field change that inhibit convection. Sunspots typically seem in pairs of conflicting magnetic polarity. This dataset is usually used to find Maxima, peak analysis, and anomaly detection in predicting an event. Peaks in the signal seem to appear at regular intervals. We Consider peaks that drop off on both sides by at least 70 relative sunspot numbers before encountering a more significant value. Figure 5 shows the predicted high-order chaotic signal with the proposed model in Sunspot dataset.

Table-2: Applied proposed model on Sunspot time-series

Evaluate Ratio	Noiseless	σN = 0.01	σN = 0.05	σN = 0.1
RMSE	0.069	0.051	0.067	0.091
rRMSE	0.096	1.081	2.267	4.089
MAE	1.653	1.092	1.791	2.171
p-Value	0.034	0.086	0.129	0.001

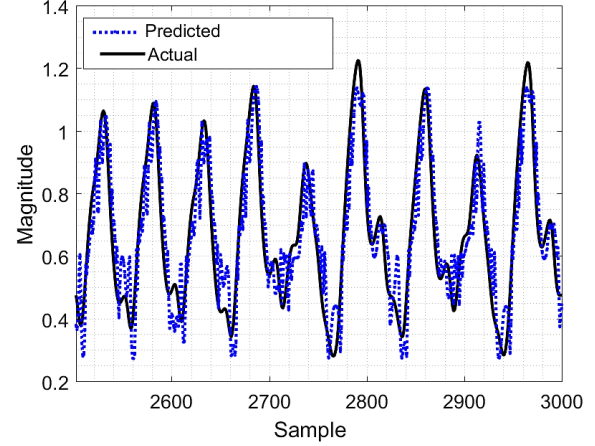


Fig 5- Predicted chaotic signal with the proposed model

Case Study 3: Football Match Time-Series

The applied training dataset contained the results of 10 seasons of the English Premier League (EPL). The dataset set is available online on the following link: <https://datahub.io/sports-data/english-premier-league>. The testing dataset included the results of two seasons of the EPL. In this reseasrech we used the form to measure the "streakiness" of a team; for example, if a team is on a hot streak, they will likely continue that hot streak.

$$GoalScore_t = ght \cdot kgh + gat \cdot kga - cht \cdot kch - cat \cdot kca$$

$$Each\ Game\ Score_t = GoalScore_t - cht - ght$$

$$TotalScore_t = f_{score} \text{ for each game}$$

$$f_{score} = \begin{cases} \text{home win, if } Score_t > \text{Home win threshold} \\ \text{away win, if } Score_t < \text{Away win threshold} \\ \text{draw, otherwise} \end{cases}$$

Table-3 Notions of the mathematical model

Notation	Description
$GoalScore_t$	Goal score for team t
gh_t	Home goals for team t
kgh	Home goal rate
ga_t	Away goals for team t
kga	Away goal rate
ch_t	Goals conceded home for team t
kch	Goal conceded home rate
ca_t	Goals conceded away for team t
kca	Goal conceded away rate
$TotalScore_t$	Total score for team t

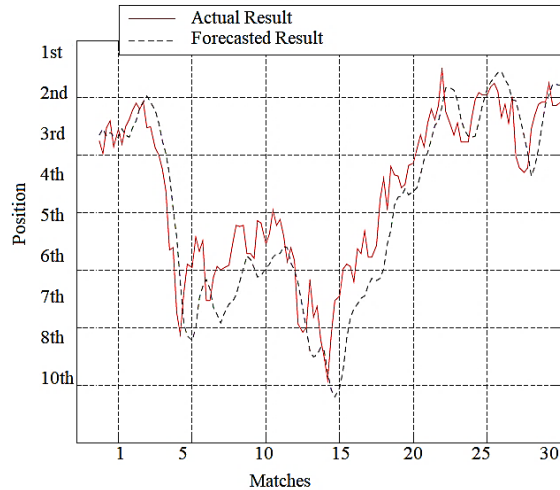


Fig.6 Performance of the proposed model based on Football Match

Figure 6 shows the prediction of real-time time series and the intensity of each value as the actual output expected during the validation phase.

Results Analysis

In Table 4, the performance evaluation of the proposed model is shown. The proposed model runs with different data sets and scenarios, including Mackey-Glass, EPL, and the Sunspot time series. The obtained results show that the FDLSTM model achieves state-of-the-art performance for global benchmarks for chaotic time-series datasets. These results show an important difference between the proposed model and its counterpart. The details of the obtained results in all three datasets are shown in Table 5. The experiments reveal the proposed model's performance in different chaotic time-series datasets.

Table-4 Performance evaluation reported in AUC%

No	EPL	Mackey Glass	Sunspot
1	90.78%	95.67%	91.33%
2	90.65%	92.29%	92.09%
3	90.12%	96.18%	96.20%
4	91.19%	93.98%	90.28%
5	92.89%	94.93%	95.47%
6	91.93%	94.56%	93.90%
7	92.89%	95.68%	94.48%
8	90.18%	94.89%	93.41%
9	93.09%	93.87%	92.04%
10	94.87%	93.23%	91.24%
Average	91.85%	94.73%	95.74%

Table -5 Performance comparison of proposed model

Dataset	AUC%	CI	Recall	Precision
EPL	91.85	[90-94] %	67%	69%
Mackey-Glass	94.73	[92-96] %	89%	91%
Sunspot	95.74	[90-96] %	91%	92%

Table-6 Comparison of state-of-the-arts with proposed model

Model	AUC%
Fuzzy Markov Chain [7]	90.79
Fractal Dimension and Fuzzy Logic [8]	91.00
Type-2 Fuzzy Time Series [9]	93.81
SEIRD with ARIMA [10]	92.29
Asymmetric Time Series [11]	93.71
Type-2 Fuzzy Time Warping [12]	94.00
FD-LSTM (average performance of this work)	94.10

Conclusion

This research has motivated and formulated to solve the problem of complex and uncertain time-series prediction in an FD-LSTM architecture. So that the constancy degree related to the complete fuzzy partiality relations obtained is as higher as possible added to the elementary structure of LSTM. Another remarkable conclusion is that FD-LSTM architectures achieve state-of-the-art performance for uncertain time-series prediction problems. The proposed architecture outperforms basic LSTM, LSTMP, and Deep-LSTM networks. It makes more effective use of the model parameters by addressing the computational efficiency and handling the associate uncertainty for uncertain and chaotic time series in short and long ranges. The obtained results with the uncertain time-series problems are revealed in the experiments. The proposed model is proficient in coping with uncertainty in uncertain conditions such as Football match prediction. The FD-LSTM architecture was designed precisely so that both short and long-term memorization and recall can happen naturally through the memory units. Hence, it is a model that is appropriate for prediction tasks under high-order uncertainty.

References

- [1] Encheng Zhang, Degang Wang, Hongxing Li, A Comprehensive High Order Type-2 Fuzzy Time-series Prediction Model, Control and Decision Conference, CCDC, Chinese, 2016.
- [2] Fernando Gaxiola, Patricia Melin, Fevrier Valdez and Juan R. Castro, Ensemble Neural Network with Type-2 Fuzzy Weights Using Response Integration for Time-series Prediction, Springer, Studies in Fuzziness and Soft Computing book series (STUDFUZZ), Vol 361, 2018.
- [3] A. Safari, R. Hosseini, M. Mazinani, "A novel deep interval type-2 fuzzy LSTM (DIT2FLSTM) model applied to COVID-19 pandemic time-series prediction", Journal of Biomedical Informatics, Vol. 123, 2021.
- [4] Keshtkar Langaroudi, M., Yamaghani, M., "Sports Result Prediction Based on Machine Learning and Computational Intelligence Approaches: A Survey". Journal of Advances in Computer Engineering and Technology, 5(1), 27-36, 2019.
- [5] Oscar Castillo, Leticia Amador-Angulo, Juan R Castro, Mario Garcia-Valdez, A comparative study of type-1 fuzzy logic systems, interval type-2 fuzzy logic systems and generalized type-2 fuzzy

logic systems in control problems, Information Sciences, vol 354, pp 257-274, 2016.

[6] Jesus Soto, Patricia Melin, Oscar Castillo, New Approach for Time-series Prediction Using Ensembles of IT2FNN Models with Optimization of Fuzzy Integrators, Taiwan Fuzzy Systems Association and Springer-Verlag GmbH Germany, part of Springer Nature 2018.

[7] K. Ramadani and D. Devianto, "The forecasting model of bitcoin price with fuzzy time series markov chain and chen logical method," in Proc. AIP Conf. 2020, vol. 2296.

[8] O. Castillo, P. Melin, "Forecasting of COVID-19 time series for countries in the world based on a hybrid approach combining the fractal dimension and fuzzy logic", Chaos Solitons Fractals, Vol. 140, pp110-142, 2020.

[9] A. Safari, R. Hosseini and M. Mazinani, "A Type-2 Fuzzy Time Series Model for Pattern Similarity Analysis: A Case Study on Air Quality Forecasting," in IEEE Intelligent Systems, doi: 10.1109/MIS.2021.3095727.

[10] M. Ala'raj, M. Majdalawieh, N. Nizamuddin, Modeling and forecasting of COVID-19 using a hybrid dynamic model based on SEIRD with ARIMA corrections, Infectious Disease Modelling, Vol. 6, pp.98-111, 2021.

[11] M. Maleki, H. Bidram, D. Wraith, "Robust clustering of COVID-19 cases across U.S. counties using mixtures of asymmetric time series models with time varying and freely indexed covariates", Journal of Applied Statistics, Special Issue: Statistical Perspectives on Analytics for COVID-19 Data, 2022.

[12] A. Safari, R. Hosseini and M. Mazinani, " Dynamic Type-2 Fuzzy Time Warping (DT2FTW): A Hybrid Model for Uncertain Time-Series Prediction", International Journal of Fuzzy Logic and Intelligent Systems Vol. 21, pp.338-348, 2021.

Authors Biography

Milad Keshtkar Langeroudi

He is with the Department of Computer Engineering, Lahijan Branch, Islamic Azad University, Lahijan, Iran. He received his MSc in artificial intelligence from Qazvin Branch in 2013. His research Interests are Pattern recognition, Deep learning, and time-series prediction.

Mohammad Reza Yamaghani

He is with the Department of Computer Engineering, Lahijan Branch, Islamic Azad University, Lahijan, Iran. He received his Ph.D. in artificial intelligence from Science and Research Branch in 2019. His research interests are Pattern recognition, Deep Neural networks, and machine vision.

Siavash Khodaparast

He is with the Department of Physical Education, Lahijan Branch, Islamic Azad University, Lahijan, Iran. He received his Ph.D. in Physical Education from Islamic Azad University in 2017. His research interests are Physical Education and Human-Activity pattern analysis